

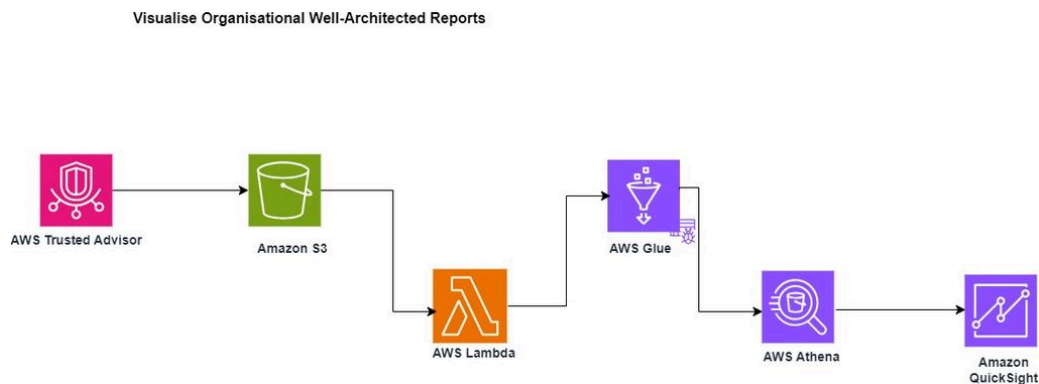
# Visualise AWS Organisation Well-Architected Reports

As Kaizen's AWS usage grows, it's essential to obtain a consolidated view of all recommendations across Kaizen accounts within the AWS Organization. These reports will be used to extract actionable insights, analyze trends, consolidate data, and provide tactical and strategic remediation as needed.

To streamline this process, we will leverage an Amazon QuickSight dashboard for seamless visualization of consolidated Trusted Advisor reports. This will enhance collaboration between teams, enabling effective remediation to optimize cost efficiency, strengthen security, and improve application performance across all AWS accounts.

## Architecture

We can create an Amazon Simple Storage Service (Amazon S3) bucket to store the reports and use an AWS Cloudformation template to create the resources described below. Then use Amazon Athena to analyse or run queries on the data generated or use Amazon QuickSight to visualise the data in a customised dashboard.



## Prerequisites and Setup Instructions [↗](#)

### Prerequisites [↗](#)

1. **IAM Permissions:** Sign in with an AWS Identity and Access Management (IAM) user with administrator permissions.
2. **AWS Region:** Use the **eu-west-1** Region for quick setup.
3. **QuickSight Account:** Create an Amazon QuickSight account. See the [Amazon QuickSight User Guide](#) for details.

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### Step 1: Upload the Report to Amazon S3 [↗](#)

1. Download the `resources-0.json`, `resources-1.json`, `resources-2.json` etc. reports.
2. **Upload to S3:**
  - Sign in to the AWS Management Console.
  - Choose **eu-west-1** in the Region selector.
  - Open the **Amazon S3 Console**.

- Select an existing bucket or create a new one.
  - Create a folder (e.g., `extracted` ) and upload the `resources.json` file to it.
  - **Note:** Rename subsequent files (e.g., `resources-timestamp.json` ) to avoid overwriting.
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## Step 2: Create Resources Using AWS CloudFormation [↗](#)

1. Use attached CloudFormation template ( AWS glue will run some aggregation of the files)
  2. Enter relevant environmental variables on deployment e.g. BucketName, FolderName (prefixes)
  3. **Create the Stack:**
    - Open the **AWS CloudFormation Console**.
    - Select **eu-west-1** in the Region selector.
    - Choose **Create stack** → **With new resources (standard)**.
    - Upload the modified YAML file as the template.
    - Enter a stack name (e.g., `Organizational-view-Trusted-Advisor-reports` ) and proceed with default options.
    - Review the settings, acknowledge IAM resource creation, and click **Create stack**.
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## Completion [↗](#)

- The stack creation takes ~5 minutes. Once complete, the **Resources** tab in the CloudFormation console will display the created resources.

## Query Data in Amazon Athena [↗](#)

1. **Set Up Athena:**
  - Open the **Athena Console** in the **eu-west-1** Region.
  - Specify a query result location in a different S3 bucket
2. **Run a Query:**
  - Navigate to **Saved Queries**, search for "Show sample."
  - Select the query (e.g., `Show sample entries of TA report` ) and run it.
  - Example Query:

```
1 sql
```

Copy code

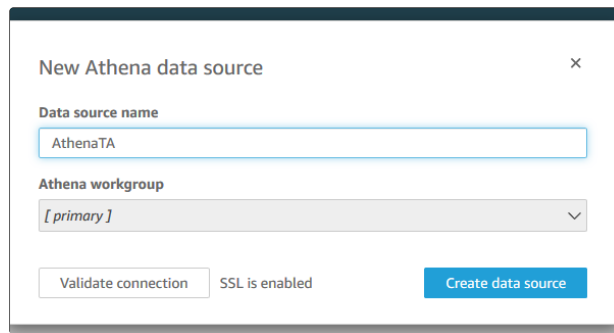
```
SELECT * FROM "athenatacfn"."extracted" LIMIT 1000000;
```

- View the query results to analyze the Trusted Advisor report.
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## Create a Dashboard in Amazon QuickSight [↗](#)

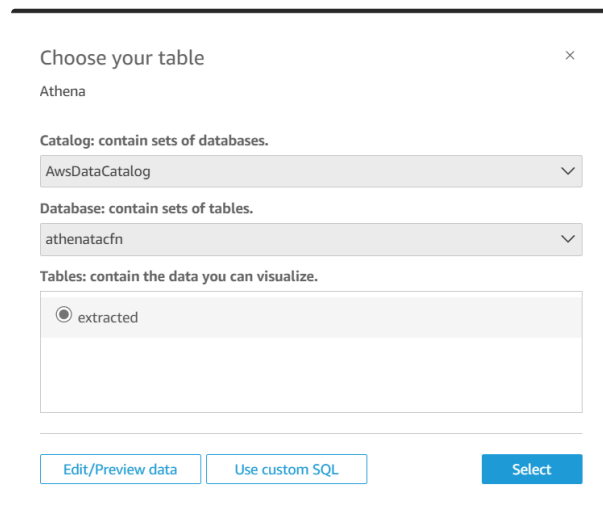
- **Set Up a Dataset:**
  - Open the **Amazon QuickSight Console** in **eu-west-1** and sign in.
  - Choose **New analysis** → **New dataset** → **Athena**.

- Enter a data source name (e.g., AthenaTA ) and create the data source.

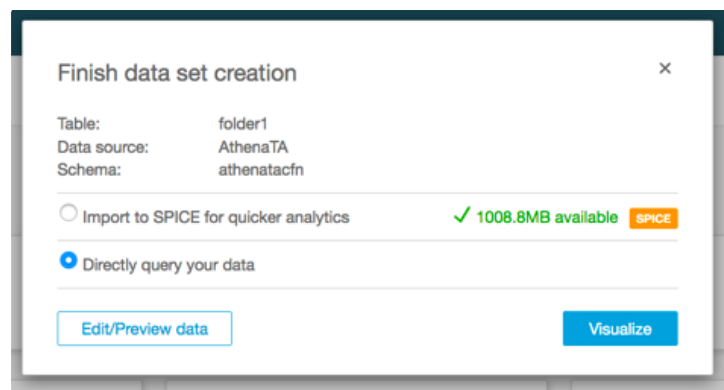


The screenshot shows a dialog box titled "New Athena data source" with a close button (X) in the top right corner. It contains two input fields: "Data source name" with the value "AthenaTA" and "Athena workgroup" with the value "[ primary ]". Below these fields are two buttons: "Validate connection" and "Create data source". A status indicator "SSL is enabled" is located between the two buttons.

- Select the athenatacfn table and the extracted dataset.



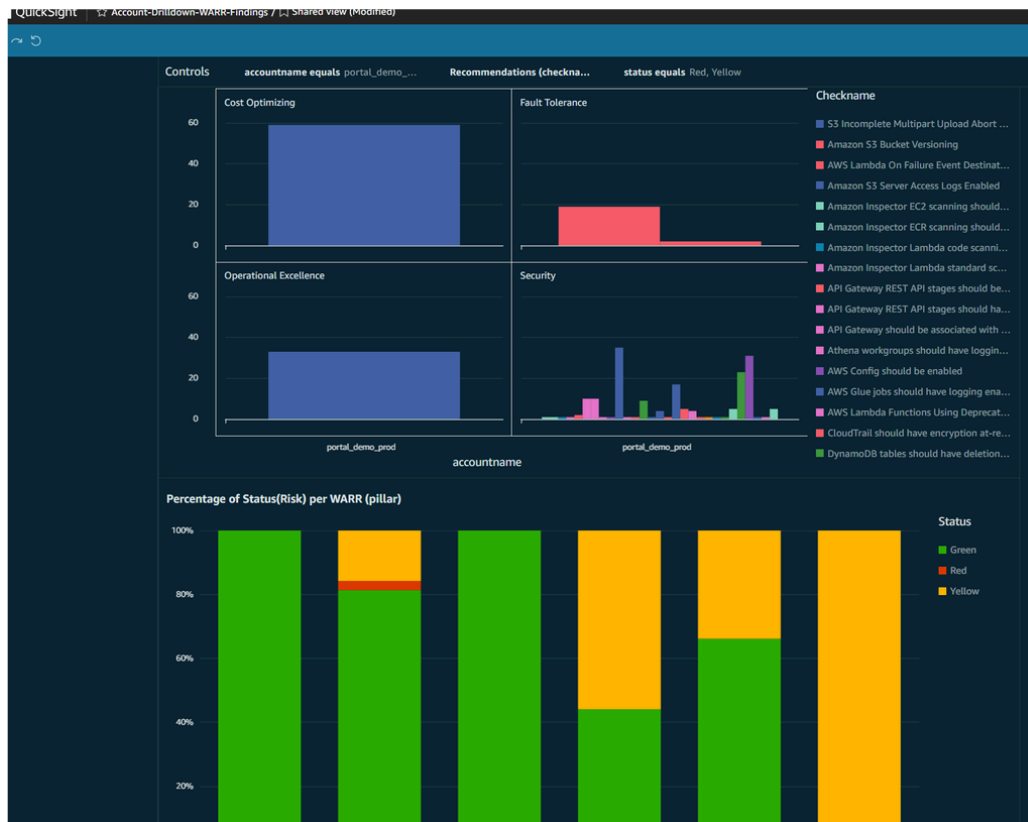
The screenshot shows a dialog box titled "Choose your table" with a close button (X) in the top right corner. It displays the hierarchy: "Athena" > "Catalog: contain sets of databases." > "AwsDataCatalog" > "Database: contain sets of tables." > "athenatacfn" > "Tables: contain the data you can visualize." > "extracted". The "extracted" table is selected with a radio button. At the bottom, there are three buttons: "Edit/Preview data", "Use custom SQL", and "Select".



The screenshot shows a dialog box titled "Finish data set creation" with a close button (X) in the top right corner. It displays the following information: "Table: folder1", "Data source: AthenaTA", and "Schema: athenatacfn". Below this, there are two options: "Import to SPICE for quicker analytics" (which is selected with a radio button) and "Directly query your data" (which is unselected). The "Import to SPICE" option has a green checkmark and the text "1008.8MB available" next to it, and a "SPICE" button. At the bottom, there are two buttons: "Edit/Preview data" and "Visualize".

- **Visualize Data:**
  - Choose **Directly query your data**, then select **Visualize**.

- Create a dashboard to view key insights, such as:
  - Affected account IDs.
  - Summary by AWS Region.
  - Check categories and statuses.
  - Number of flagged entries per account.



For more details, see [Amazon QuickSight Dashboards](#).